

Vaishnavi Tripathi

DATA ANALYST | SQL | PYTHON | MACHINE LEARNING | POWER BI

Melbourne VIC | Australian Permanent Resident – Full Work Rights

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Professional Summary

Data Analyst experienced in transforming education and digital platform data into actionable insights using SQL, Python, and Power BI. Experienced in conversion analysis, KPI development, data modelling and database-driven applications, with a track record of translating student and commercial data into measurable business outcomes.

Technical Skills

- **Data Analysis & Programming:** SQL, Python, pandas, NumPy, Jupyter Notebook, Exploratory Data Analysis
 - **Machine Learning & Deep Learning:** scikit-learn, Data Pre-processing, Feature Engineering, Classification, Regression, Predictive Modelling, Model Evaluation, Neural Networks, Convolutional Neural Networks (CNNs).
 - **Databases & Data Platforms:** MySQL, MongoDB, Relational Databases, NoSQL, Snowflake Fundamentals.
 - **Business Intelligence:** Power BI, DAX, Power Query, Data Modelling, KPI Development, Dashboard Design, Data Visualisation
 - **Data Management & Integration:** Data Cleaning, Transformation, Validation, Reconciliation, Data Quality, Data Mapping, ETL/ELT Concepts, API Integration
 - **Generative AI:** Large Language Models (LLMs), Generative AI Fundamentals, Prompt Engineering Fundamentals
 - **Enterprise & Development Tools:** SAP, Salesforce, Git, Jira, Xero
- Supporting Tools:** Microsoft Excel, Microsoft Office

Employment Experience

Data Analyst

Nov 2025 - Present

Stepping Stone | Melbourne, Australia

Role Summary: Analyse student behaviour and commercial performance across an EdTech platform, while building PickMyUni — a data-driven university comparison tool.

Data Analytics & Business Intelligence

- Query student enquiry, enrollment, course, fee, and revenue data to surface conversion trends and course-selection patterns driving business decisions.
- Clean, transform, validate and reconcile datasets to improve data quality, consistency and analytical reliability.
- Built Power BI data models and DAX measures tracking inquiry-to-enrollment conversion, revenue, retention, and course performance across KPIs.
- Analyzed student preferences, course migration, pricing, and withdrawal behavior to recommend course planning, pricing, and retention strategies adopted by leadership.
- Delivered insights supporting pricing and retention initiatives, contributing to a 22% increase in average revenue per student and a 30% reduction in course withdrawals.

Data Platform & Website Development – PickMyUni

- Designed a relational database in MySQL to structure university, course, and fee data for efficient search and comparison across Australian institutions.

- Developed SQL queries and data structures supporting course search, filtering and comparison functionality across multiple institutions.
- Integrated APIs from RMIT University and Deakin University to pull live university and course data into the platform.
- Transformed and standardized data from multiple APIs into consistent MySQL schemas, enabling course comparison and front-end presentation for students.

Data Analyst

Jul 2025 - Nov 2025

Saras Care - Melbourne Diwali App | Melbourne, Australia

Role Summary: Supported data operations for a digital event platform using SQL and user engagement data to assist reporting, validation and application performance analysis.

- Gathered and translated stakeholder requirements into structured data, reporting and application requirements for a digital event platform.
- Cleaned 300+ registration records in SQL, removing duplicates, incomplete entries, and test data to enable reliable conversion analysis.
- Analyzed registration and engagement data across 500+ sign-ups in SQL, surfacing a 40% registration-to-ticket conversion rate that informed capacity planning.
- Developed Power BI dashboards and analytical reports to communicate registration, engagement and operational insights.
- Queried MongoDB to analyze user submissions and feedback, surfacing data quality issues that informed backend improvements for a scalable event platform.
- Delivered analytical figures supporting operational and catering planning for 300+ attendees.

Finance & Reporting Analyst

Aug 2023 - Sep 2025

Victorian College of Education | Melbourne, Australia

Role Summary: Supported finance operations through data analysis, reconciliation and reporting across student enrolment and tuition payment records.

- Reconciled student enrollment, tuition-payment, and financial data across systems, identifying discrepancies between enrollment and payment records to improve data quality and reporting accuracy.
- Built recurring reports in Excel covering payment status, outstanding balances, and student accounts for finance and student-services teams.
- Collaborated with cross-functional stakeholders to investigate reporting issues and resolve data discrepancies.
- Used Xero and Excel as supporting tools for reconciliation, validation and ad-hoc analysis.

Associate Software Engineer

Jul 2022 - Jul 2023

Accenture | India

Role Summary: Supported SAP and Salesforce enterprise data workflows with a focus on data validation, system integration, testing and data-quality improvement.

- Developed and optimised enterprise data workflows using SAP and Salesforce.
- Built validation tools that enhanced data accuracy and reduced manual processing across integrated systems.
- Supported systems integration, troubleshooting, and issue resolution across enterprise platforms.
- Collaborated with technical teams to investigate and resolve data and system issues across SAP and Salesforce environments.
- Participated in requirements analysis, software testing, and technical documentation.
- Built validation tools that enhanced data accuracy and reduced manual processing across integrated systems.

Education

Master of Information Technology

May 2023 - Nov 2025

Victorian Institute of Technology | Melbourne, Victoria, Australia

- Completed postgraduate studies in Information Technology with emphasis on software development, enterprise systems, systems analysis, and databases

Selected Data & Machine Learning Projects

Predictive Modelling with Machine Learning

- Built end-to-end machine-learning workflows in Python on real-world datasets, covering data cleaning, exploratory data analysis and feature engineering through to prediction.
- Trained, tuned and compared predictive models using scikit-learn, evaluating with R-squared and F1-score, then interpreted results to show how the models support business decisions.

Technologies: Python | Machine Learning | Feature Engineering | Data Pre-processing | Model Evaluation

Traffic Sign Recognition & Obstacle Detection – Deep Learning / CNN Project

- Built a CNN in Python to classify 43 traffic-sign types with 95% test accuracy across 50,000 images.
- Applied data augmentation (rotation, scaling, brightness adjustment) to improve model generalisation and reduce overfitting.
- Extended the model to detect obstacles in real-time video, integrating OpenCV for live frame processing and bounding-box annotation.

Technologies: Python | CNN | Deep Learning | Computer Vision | Image Classification | Data Augmentation

Technical Research:

- Deep Learning & Computer Vision: Neural Networks, CNN Architecture, Convolution, Feature Extraction, Pooling, Image Classification, Model Training and Validation
- Modern Data Platforms & Databases: Snowflake Fundamentals, NoSQL Databases, ETL/ELT Concepts
- Generative AI & LLMs: Large Language Model Fundamentals, Generative AI Concepts, Prompt Engineering, AI-assisted Data Analytics